

1 DEFAULT DRAINED SOIL

Preliminary values for ordinary granular retained soil. Project geotechnical recommendations govern.

PARAMETER	DEFAULT	USE
Effective retained-soil friction angle, ϕ'	28°	Drained strength
Moist total unit weight, γ	120–125 pcf	Retained-soil weight
Foundation interface friction coefficient, μ	0.35	Preliminary sliding resistance
Uniform surface surcharge, q	100 psf	Minimum general surcharge
Allowable soil bearing pressure, q_a	2,000 psf	Preliminary footing sizing
Rankine active coefficient, K_a	0.361	Yielding wall
Normally consolidated at-rest coefficient, K_0	0.531	Restrained wall
Active equivalent-fluid pressure	43.3–45.1 pcf	$K_a\gamma$
At-rest equivalent-fluid pressure	63.7–66.3 pcf	$K_0\gamma$

Level backfill; vertical wall; no cohesion; no groundwater or hydrostatic pressure. Confirm μ and q_a for the foundation soil.

2 DEFAULT UNDRAINED SOIL

No wall drainage is a saturated effective-stress condition. It is not automatically a $\phi_u = 0$ total-stress condition.

SATURATED / NO-DRAIN RETAINED SOIL

PARAMETER	DEFAULT	USE
Effective retained-soil friction angle, ϕ'	28°	Saturated drained strength
Saturated total unit weight, γ_{sat}	125 pcf	Total soil weight
Fresh-water unit weight, γ_w	62.4 pcf	Hydrostatic pressure
Submerged unit weight, γ'	62.6 pcf	$\gamma_{\text{sat}} - \gamma_w$
Rankine active coefficient, K_a	0.361	Yielding wall
Normally consolidated at-rest coefficient, K_0	0.531	Restrained wall
Active total equivalent-fluid pressure	85.0 pcf	$K_a\gamma' + \gamma_w$
At-rest total equivalent-fluid pressure	95.6 pcf	$K_0\gamma' + \gamma_w$
Active lateral surcharge pressure	36.1 psf	$K_a q$
At-rest lateral surcharge pressure	53.1 psf	$K_0 q$
Foundation interface coefficient, μ	Site-specific	Include uplift in base normal force
Allowable soil bearing pressure, q_a	Site-specific	Saturated foundation soil

Groundwater at retained grade. Effective soil pressure and full hydrostatic pressure are included in the total equivalent-fluid pressures above.

TRUE UNDRAINED TOTAL-STRESS SOIL

PARAMETER	DEFAULT	USE
Total-stress friction angle, ϕ_u	0°	Saturated undrained clay
Saturated total unit weight, γ_{sat}	125 pcf	Total-stress soil weight
Undrained shear strength, s_u	Site-specific	Required geotechnical input
Foundation interface adhesion, c_a	Site-specific	Preliminary sliding resistance
Uniform surface surcharge, q	100 psf	Minimum general surcharge
Allowable soil bearing pressure, q_a	Site-specific	Determine from foundation-soil s_u

For a yielding wall with level backfill:

$$p_a(z) = \max[0, \gamma_{\text{sat}}z + q - 2s_u] \quad (1)$$

$p_a(z)$ total active lateral pressure at depth z , psf
 z depth below retained grade, ft

Do not derive s_u from ϕ' , μ , or q_a . Do not add pore pressure separately throughout a consistent total-stress analysis; account separately for free water, differential water levels, uplift, and water-filled tension cracks.